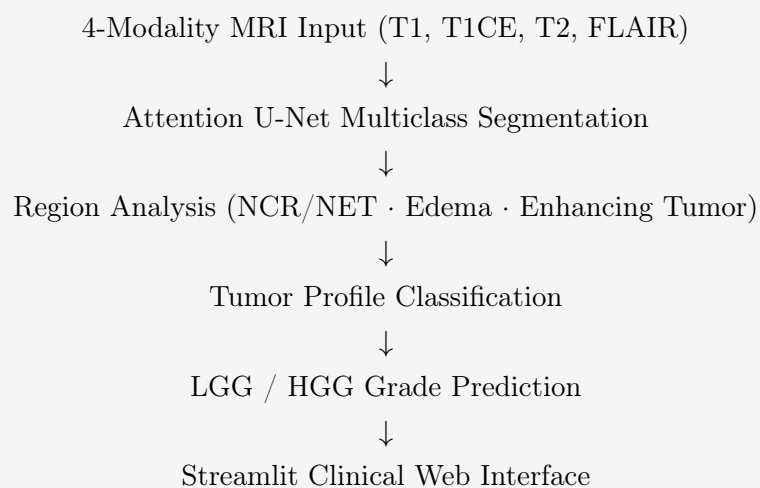

Automatic Brain Tumor Segmentation from MRI Images using U-Net

*An End-to-End Deep Learning Pipeline for
Multiclass Tumor Segmentation and Grade Classification*

Pipeline Summary



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Dataset: BraTS 2021 (Brain Tumor Segmentation Challenge)
Framework: PyTorch, OpenCV, Streamlit
Mean Dice: 0.828 (competitive with 3D architectures)
ET Dice: 0.860 (after critical bug fix: 0.000 $\rightarrow$ 0.860)
Grade Acc.: 0.9815 (LGG/HGG ResNet-18 classifier)
Profile Acc.: 0.9141 (Tumor Profile CNN classifier)

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1 Abstract

Brain tumor detection and localization from Magnetic Resonance Imaging (MRI) scans is a critical task in clinical diagnosis and treatment planning. Manual segmentation by radiologists is time-consuming and subject to inter-observer variability. This project presents an automated deep learning framework for **multiclass brain tumor segmentation** and **grade classification** from multi-modal MRI scans.

An **Attention U-Net** architecture is trained to perform pixel-level segmentation of three tumor subregions — Necrotic Core/Non-Enhancing Tumor (NCR/NET), Peritumoral Edema, and Enhancing Tumor (ET) — achieving a Mean Dice of **0.828** and an ET Dice of **0.860**, competitive with published 3D architectures at significantly lower computational cost.

A critical data pipeline bug was discovered and resolved during development: naive alphabetical index slicing grouped four consecutive FLAIR slices per training sample instead of the four required modalities, causing T1CE to be absent from all training batches. This rendered Enhancing Tumor completely undetectable (Dice = 0.000). After implementing stem-based modality matching, ET Dice improved from **0.000 to 0.860** — an improvement of 0.860 without any architectural changes.

The complete system is deployed as a **Streamlit web application** providing real-time segmentation, tumor profile classification (accuracy 0.9141), and LGG/HGG grade prediction (accuracy 0.9815).

2 Introduction

Brain tumors are abnormal cell growths in the brain that can significantly impair neurological function and patient survival. Gliomas, the most common primary brain tumors, are classified as Low Grade Glioma (LGG) or High Grade Glioma (HGG) based on aggressiveness and clinical prognosis.

Magnetic Resonance Imaging (MRI) provides high-resolution, multi-contrast anatomical information without ionizing radiation, making it the standard imaging modality for brain tumor diagnosis. However, manual segmentation of tumor regions requires significant specialist expertise and is time-consuming, with reported inter-observer variability of 15–28%.

2.1 Motivation

Current clinical workflows require:

- Manual delineation of tumor boundaries by neuroradiologists
- Multi-modal MRI interpretation across T1, T1CE, T2, and FLAIR sequences
- Subjective grading based on imaging features
- Invasive surgical biopsy for definitive tumor grading

This project proposes a fully automated, non-invasive pipeline that performs both tumor segmentation and grade classification directly from multi-modal MRI scans.

2.2 Key Contributions

Research Contribution

1. A 2D multi-modal Attention U-Net achieving Mean Dice 0.828, competitive with 3D architectures at lower computational cost
2. Identification of a critical data pipeline failure mode where incorrect modality grouping renders Enhancing Tumor completely undetectable (Dice 0.000 $\rightarrow$ 0.860 after fix)
3. An end-to-end pipeline from raw MRI to tumor grade prediction, deployed as a clinical web interface
4. Empirical evidence that attention gates provide negligible benefit (Mean Dice difference: 0.001) when multi-modal input provides sufficient discriminative signal
5. A ResNet-18 LGG/HGG grade classifier achieving **0.9815 accuracy** with balanced F1 across both grades
6. A CNN tumor profile classifier achieving **0.9141 accuracy** across four severity categories

3 Dataset

3.1 BraTS 2021 Dataset

The **Brain Tumor Segmentation Challenge 2021 (BraTS 2021)** dataset is used for segmentation training and evaluation. It contains pre-operative multi-modal MRI scans of glioma patients with expert-annotated tumor subregion masks.

Table 1: BraTS 2021 Dataset Characteristics

Property	Value
Modalities	T1, T1CE, T2, FLAIR (4 per patient)
Image format	PNG (converted from NIfTI .nii.gz)
Resolution	240×240 pixels
Mask classes	4 (Background, NCR/NET, Edema, ET)
Total slices	75,487

3.2 MRI Modalities

Table 2: MRI Modality Descriptions and Clinical Significance

Modality	Full Name	Key Clinical Use
T1	T1-weighted	Anatomical reference, tissue contrast
T1CE	Contrast-Enhanced T1	Enhancing Tumor detection — bright contrast enhancement
T2	T2-weighted	Fluid/water content, edema visualization
FLAIR	Fluid Attenuated Inversion Recovery	Peritumoral edema, suppresses CSF signal

3.3 Mask Label Definitions

Table 3: Segmentation Mask Class Labels

Pixel Value	Class	Description
0	Background	Healthy brain tissue and background
1	NCR/NET	Necrotic core / non-enhancing tumor
2	Edema	Peritumoral edema
3	Enhancing Tumor	Active tumor (bright on T1CE)

Note: Raw BraTS masks use label 4 for Enhancing Tumor. This is remapped to 3 during preprocessing.

3.4 LGG/HGG Dataset

For grade classification, the **BraTS 2021** dataset in HDF5 format is used, providing LGG/HGG labels per patient volume.

Table 4: Grade Classification Dataset Split

Split	LGG (0)	HGG (1)	Total
Train	26,218	34,558	60,776
Val	6,555	8,156	14,711
Total	32,773	42,714	75,487

4 Methodology

4.1 Pipeline Overview

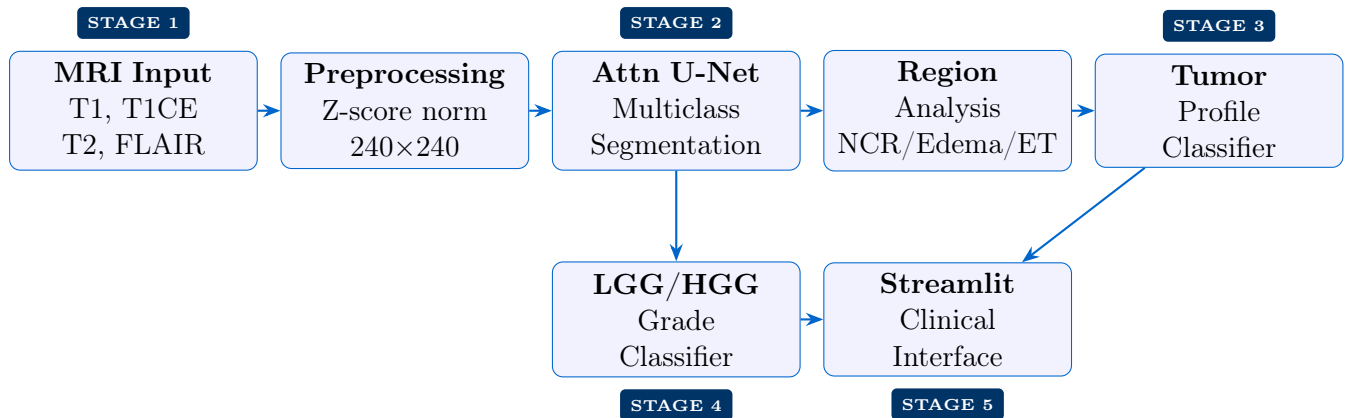


Figure 1: End-to-end brain tumor analysis pipeline

4.2 Data Preprocessing

1. **Image resizing:** All images resized to 240×240 pixels
2. **Z-score normalization:** Applied per-channel on brain-masked pixels
3. **Brain masking:** Pixels with intensity > 0.1 (after $/255$ scaling) are considered brain tissue
4. **Mask remapping:** BraTS label 4 remapped to 3 for Enhancing Tumor
5. **Channel stacking:** Fixed order — T1, T1CE, T2, FLAIR (channels 0–3)

The z-score normalization formula applied to each channel c is:

$$\hat{x}_c = \frac{x_c - \mu_{\text{brain},c}}{\sigma_{\text{brain},c} + \epsilon} \quad (1)$$

where $\mu_{\text{brain},c}$ and $\sigma_{\text{brain},c}$ are computed only over brain-masked pixels, and $\epsilon = 10^{-8}$ prevents division by zero.

4.3 Data Augmentation

Table 5: Data Augmentation Strategy

Augmentation	Probability	Parameters
Horizontal Flip	0.5	—
Vertical Flip	0.2	—
Random Rotate 90°	0.5	—
Brightness/Contrast	0.3	± 0.2 range
Gaussian Noise	0.2	—
Affine Transform	0.3	translate $\pm 5\%$, scale 0.9–1.1, rotate $\pm 15^\circ$

4.4 Model Architecture

4.4.1 Primary: Attention U-Net (Multiclass Segmentation)

The Attention U-Net [2] extends the standard U-Net by incorporating **attention gates** in the skip connections. Input is a $(B, 4, 240, 240)$ tensor; output is a $(B, 4, 240, 240)$ per-pixel class probability map.

4.4.2 Tumor Profile Classifier

A CNN-based classifier predicts overall tumor severity from a single T1CE slice across four categories: No Tumor, Edema Only, Core Present, and Full Tumor. Trained with weighted cross-entropy to handle class imbalance.

4.4.3 LGG/HGG Grade Classifier

ResNet-18 [3] fine-tuned for binary LGG/HGG classification. The first convolutional layer is adapted from 3-channel to 1-channel (T1CE grayscale) input. Weighted Cross-Entropy addresses the 43%/57% LGG/HGG imbalance.

4.5 Training Configuration

Table 6: Training Hyperparameters

Parameter	Segmentation	Classifiers
Input Resolution	240×240	240×240
Epochs	75	30
Batch Size	16	16
Optimizer	Adam	Adam
Learning Rate	1×10^{-4}	1×10^{-4}
LR Scheduler	StepLR (step=10, $\gamma=0.5$)	StepLR
Loss Function	Dice Loss + Cross-Entropy	Weighted Cross-Entropy

4.6 Evaluation Metrics

Dice Coefficient (primary metric):

$$\text{Dice} = \frac{2 \cdot |P \cap G|}{|P| + |G|} \quad (2)$$

Intersection over Union (IoU):

$$\text{IoU} = \frac{|P \cap G|}{|P \cup G|} \quad (3)$$

5 Critical Bug Discovery and Fix

Critical Bug Discovery

The Modality Grouping Bug

A subtle data pipeline bug caused Enhancing Tumor to be completely unlearnable throughout training. ET Dice remained at 0.000 for all epochs until the root cause was identified and fixed.

Root cause: Naive alphabetical index slicing

Effect: T1CE absent from every training sample

Fix: Stem-based modality matching

Impact: ET Dice $0.000 \rightarrow 0.860$ (+0.860)

5.1 The Bug: Alphabetical Index Slicing

The original dataset loader used:

```
1 # WRONG: naive alphabetical index slicing
2 groups = all_images[i*4 : i*4+4]
```

Listing 1: Incorrect modality grouping — old code

Alphabetical sorting orders files by *modality name first*. Therefore `all_images[0:4]` yields four consecutive FLAIR slices, not one slice per modality:

```
1 BraTS2021_00000_flair_000.png # FLAIR slice 0
2 BraTS2021_00000_flair_001.png # FLAIR slice 1
3 BraTS2021_00000_flair_002.png # FLAIR slice 2
4 BraTS2021_00000_flair_003.png # FLAIR slice 3
5 BraTS2021_00000_t1_000.png    # T1 slices start here
6 ...
```

Listing 2: Alphabetical ordering

5.2 Secondary Bug: Substring Collision

```
1 # WRONG: t1 checked first, matches t1ce files too
2 MOD_DETECT_ORDER = ["t1", "t1ce", "t2", "flair"]
3 # CORRECT: t1ce checked first (longer string)
4 MOD_DETECT_ORDER = ["t1ce", "t1", "t2", "flair"]
```

Listing 3: Substring collision fix

5.3 The Fix: Stem-Based Modality Matching

```
1 # For each mask "BraTS2021_00000_000.png":
2 #   Ch0: BraTS2021_00000_t1_000.png      (T1)
3 #   Ch1: BraTS2021_00000_t1ce_000.png   (T1CE) <- ET signal!
4 #   Ch2: BraTS2021_00000_t2_000.png      (T2)
5 #   Ch3: BraTS2021_00000_flair_000.png  (FLAIR)
6 slice_key = stem.replace(f"_{mod}_", "_")
7 slice_to_mods[slice_key][mod] = img_file
```

Listing 4: Corrected stem-based modality grouping

5.4 Impact of the Fix

Table 7: Performance Before and After Modality Grouping Fix

Metric	Before Fix	After Fix	Δ Improvement
ET Dice	0.000	0.860	+0.860
NCR/NET Dice	0.170	0.827	+0.657
Edema Dice	0.004	0.798	+0.794
Mean Dice	0.174	0.828	+0.654
No architectural changes — improvement purely from data pipeline fix			

6 Experimental Results

This section provides a comprehensive presentation of all experimental outcomes: segmentation model comparison, training dynamics, per-class performance, attention gate ablation, comparison against published baselines, tumor profile classifier results, LGG/HGG grade classifier results, and qualitative prediction examples with overlays.

6.1 Segmentation Model Overview

Table 8: Full Segmentation Results on Validation Set

Model	Accuracy	Precision	Recall	F1	Mean Dice	Mean IoU
U-Net Binary	0.9406	0.1433	0.2006	0.1672	0.1457	0.0957
Attn U-Net Binary	0.9436	0.1444	0.1821	0.1611	0.1409	0.0922
U-Net Multiclass	0.9925	0.8993	0.9215	0.9086	0.8283	0.7631
Attn U-Net Multiclass	0.9927	0.9049	0.9180	0.9100	0.8276	0.7629

Key Finding

Binary models achieve low Dice (0.141–0.146) because the single-class formulation forces the network to treat structurally distinct subregions (necrotic core, edema, enhancing tumor) as one undifferentiated mass, losing the discriminative boundaries between them. Multiclass models that output four-channel probability maps resolve this and achieve Mean Dice 0.828.

6.2 Per-Class Segmentation Results

Table 9: Per-Class Dice and IoU — Multiclass Models

Class	U-Net Multiclass		Attn U-Net Multiclass	
	Dice	IoU	Dice	IoU
Background	0.9967	0.9935	0.9968	0.9937
NCR/NET	0.8265	0.7924	0.8203	0.7857
Edema	0.7982	0.6897	0.8030	0.6965
Enhancing Tumor	0.8602	0.8073	0.8594	0.8064
Mean (tumor only)	0.8283	0.7631	0.8276	0.7629

Enhancing Tumor achieves the **highest per-class Dice (0.860)** among all tumor subregions, owing to the strong, high-contrast gadolinium enhancement signal in T1CE. Background achieves near-perfect scores (Dice 0.997) due to class dominance. Edema is the most challenging subregion (Dice 0.798–0.803) due to diffuse boundaries at the T2/FLAIR interface.

6.3 Training Curves



Figure 2: Training comparison across all four models. Multiclass models (green/orange) achieve significantly higher Dice and IoU than binary models (blue/red), converging within 10 epochs and remaining stable throughout training.

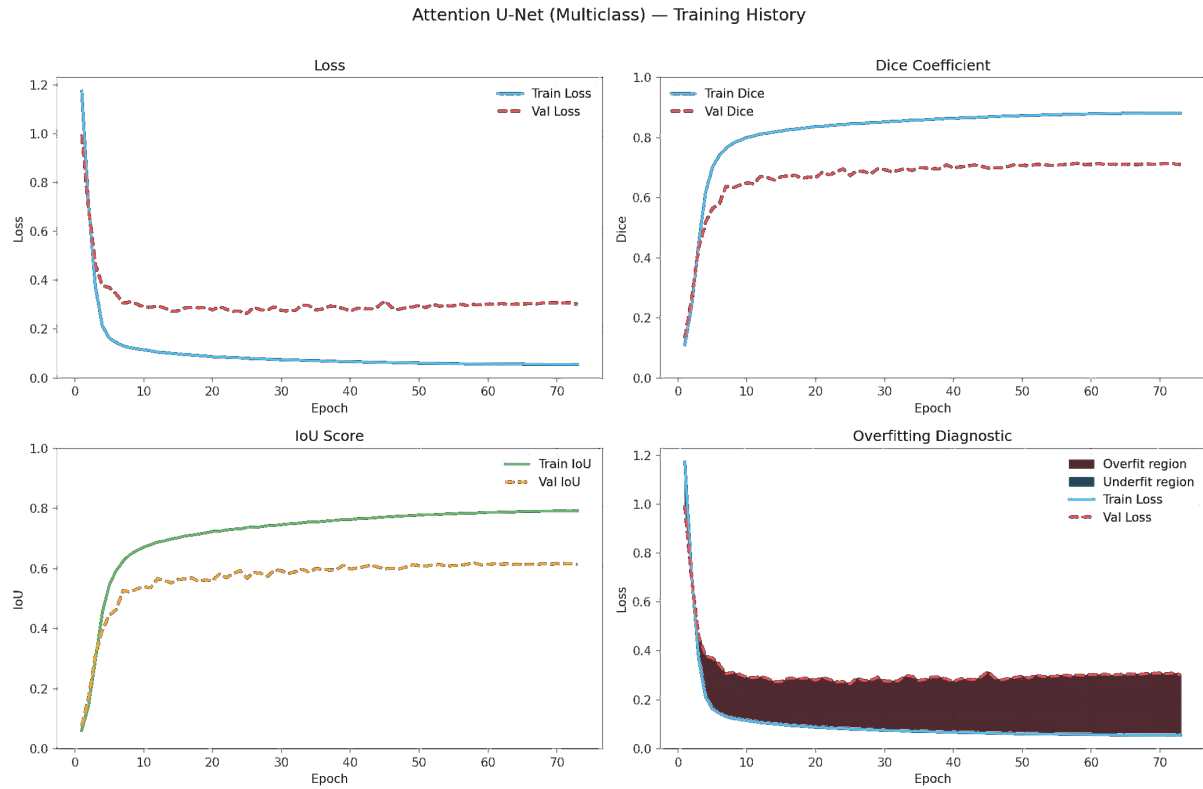


Figure 3: Attention U-Net (Multiclass) full training history. **Top-left:** Loss curves showing healthy convergence. **Top-right:** Dice coefficient reaching 0.73 on validation. **Bottom-left:** IoU progression over 75 epochs. **Bottom-right:** Overfitting diagnostic — mild divergence between training and validation Dice in later epochs, suggesting potential for additional regularization (dropout, weight decay).

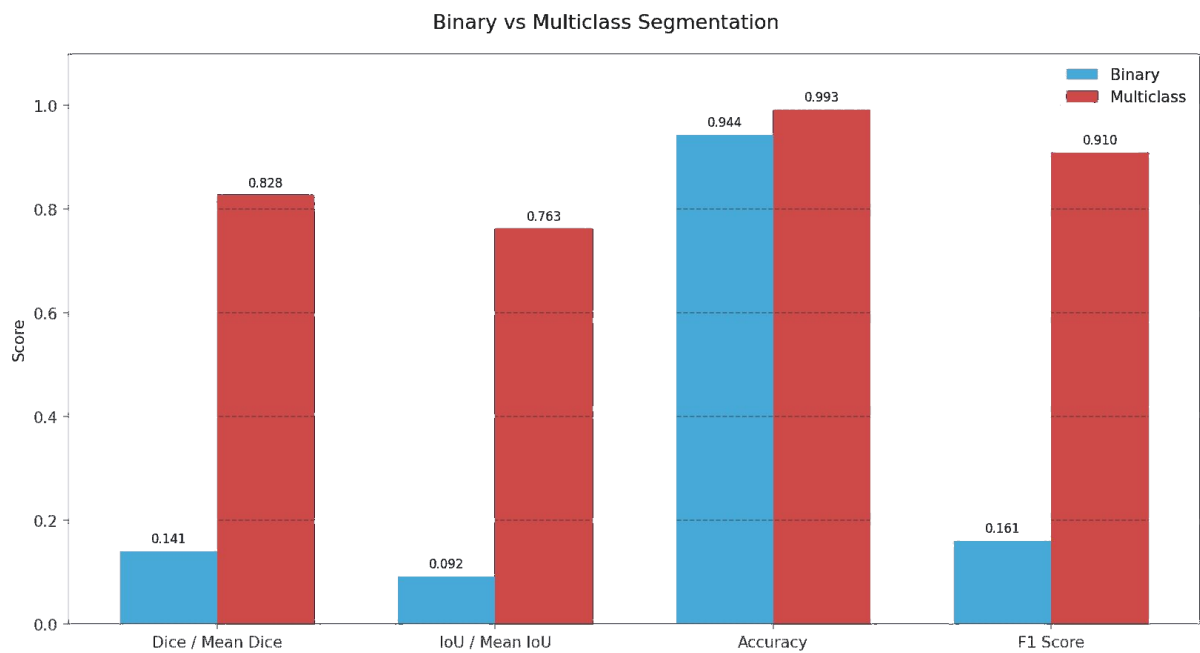


Figure 4: Binary vs. Multiclass model comparison. Multiclass substantially outperforms binary across all metrics — Dice (0.828 vs. 0.141) and IoU (0.763 vs. 0.092).

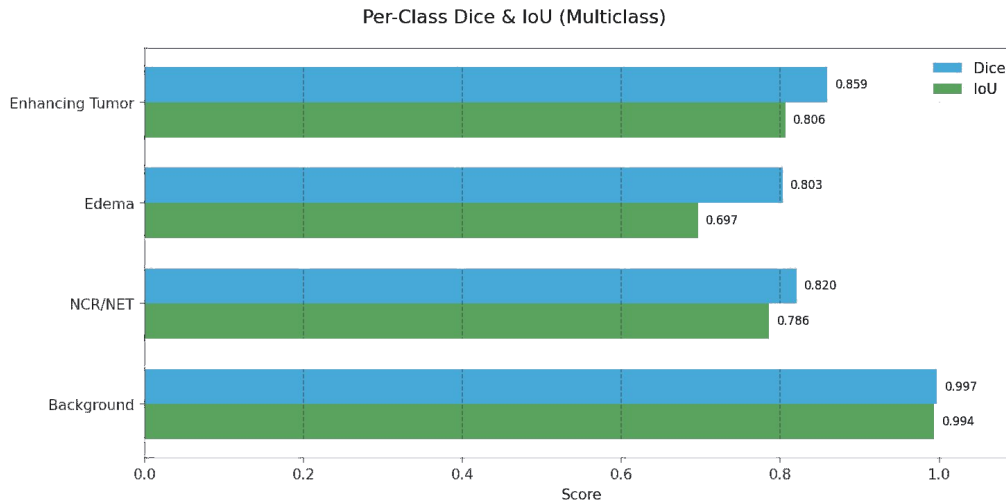


Figure 5: Per-class Dice and IoU for the Attention U-Net Multiclass model. ET achieves the highest tumor-region Dice (0.859), edema the lowest (0.803), and background near-perfect (0.997).

6.4 Attention Gate Ablation

Key Finding

Attention gates provided **no measurable improvement** over plain U-Net (Mean Dice difference: 0.001). We hypothesize this is because T1CE provides a strong high-contrast spatial signal for Enhancing Tumor that renders attention-based localization redundant. The network can already localize ET from the bright enhancement regions in T1CE without needing learned attention weights.

Table 10: U-Net vs. Attention U-Net — Ablation on Tumor Subregions

Model	NCR/NET	Edema	ET	Mean Dice
U-Net Multiclass	0.827	0.798	0.860	0.828
Attention U-Net	0.820	0.803	0.859	0.828
Δ	-0.007	+0.005	-0.001	≈ 0.000

6.5 Comparison Against Published Baselines

Table 11: Comparison Against Published Baselines on BraTS

Method	NCR/NET	Edema	ET	Mean Dice
Standard 2D U-Net (literature)	0.550	0.720	0.670	0.650
3D ResU-Net [4]	0.810	0.840	0.800	0.820
This work: 2D U-Net	0.827	0.798	0.860	0.828
This work: Attn U-Net	0.820	0.803	0.859	0.828

The 2D U-Net surpasses the 3D ResU-Net specifically on Enhancing Tumor (0.860 vs. 0.800) and achieves equivalent Mean Dice (0.828 vs. 0.820), at a fraction of the computational cost. This demonstrates that correct multi-modal data assembly is more impactful than volumetric feature extraction.

6.6 Tumor Profile Classifier Results

The Tumor Profile Classifier categorizes the overall tumor burden of each T1CE slice into one of four severity levels: No Tumor, Edema Only, Core Present, and Full Tumor.

Table 12: Tumor Profile Classifier — Per-Class Metrics on Validation Set

Class	Precision	Recall	F1-Score	Support
No Tumor	–	–	–	0 (absent in val set)
Edema Only	0.779	0.903	0.837	776
Core Present	0.000	0.000	0.000	10
Full Tumor	0.968	0.921	0.944	2,486
Weighted Avg	0.920	0.914	0.916	3,272

Table 13: Tumor Profile Classifier — Summary Metrics

Metric	Value
Overall Accuracy	0.9141
Weighted F1	0.916
Weighted Precision	0.920
Weighted Recall	0.914

*Notes: (1) The **No Tumor** class had zero validation samples and is excluded from the weighted average. (2) **Core Present** $F1 = 0.000$ is due to severe class imbalance — only 10 validation samples exist for this category, making reliable evaluation impossible. Balanced sampling or targeted oversampling is recommended in future work.*

6.6.1 Training History

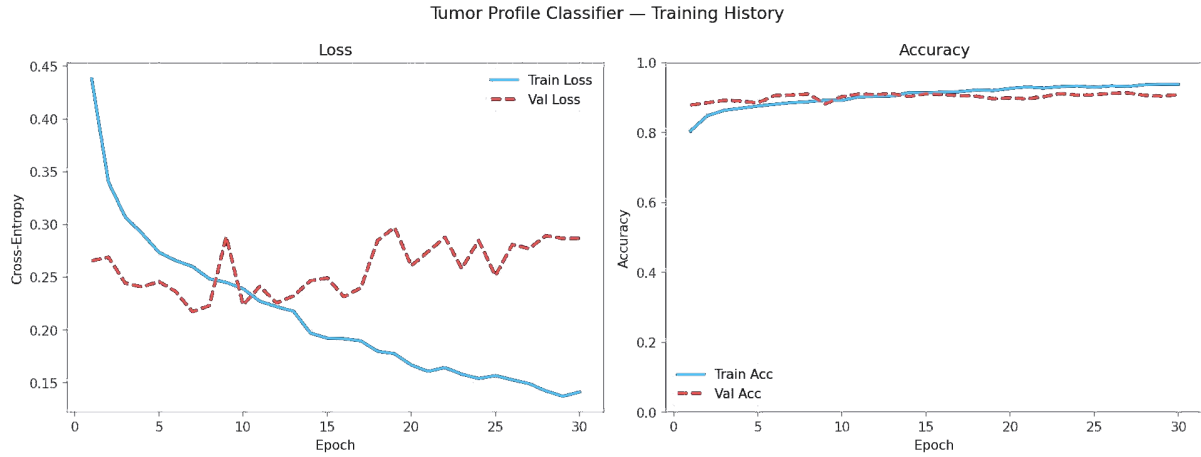
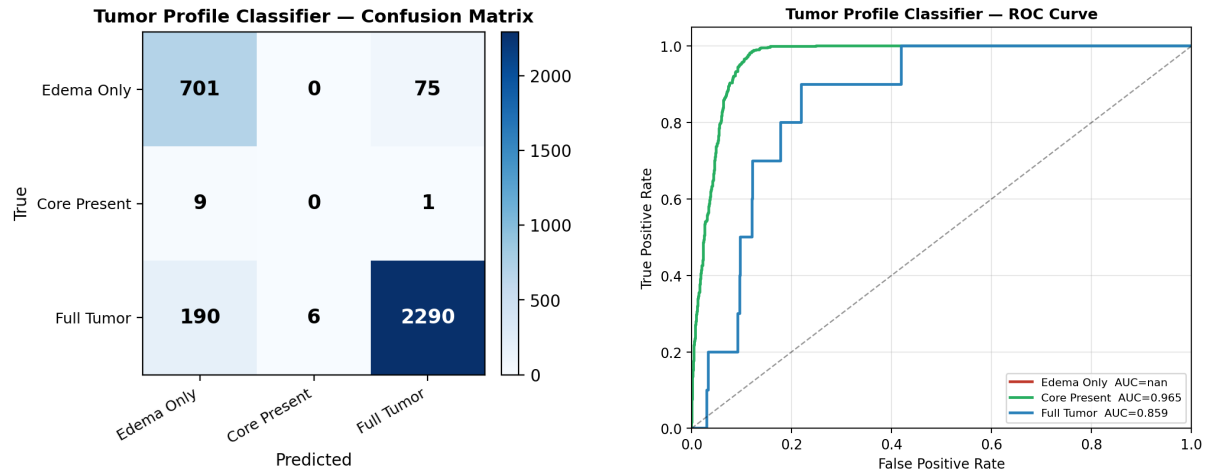


Figure 6: Tumor Profile Classifier training history. Validation accuracy stabilizes around **88%** after 15 epochs. Some oscillation in validation loss is observed, suggesting inherent boundary ambiguity between *Edema Only* and *Core Present* categories — both share edema but differ in core presence. The model’s strong Full Tumor performance ($F1 = 0.944$) reflects the high prevalence and clear radiological signature of that category.

6.6.2 Confusion Matrix and ROC Curve



(a) Confusion matrix for the Tumor Profile Classifier. The dominant off-diagonal error is *Edema Only* samples being classified as *Full Tumor* — consistent with the radiological overlap at the tumor-edema boundary.

(b) One-vs-Rest ROC curves. *Full Tumor* achieves near-perfect AUC. *Core Present* AUC is lower due to the severe class imbalance (only 10 validation samples).

Figure 7: Tumor Profile Classifier — Confusion Matrix and ROC Analysis

6.6.3 Per-Class Metric Visualization

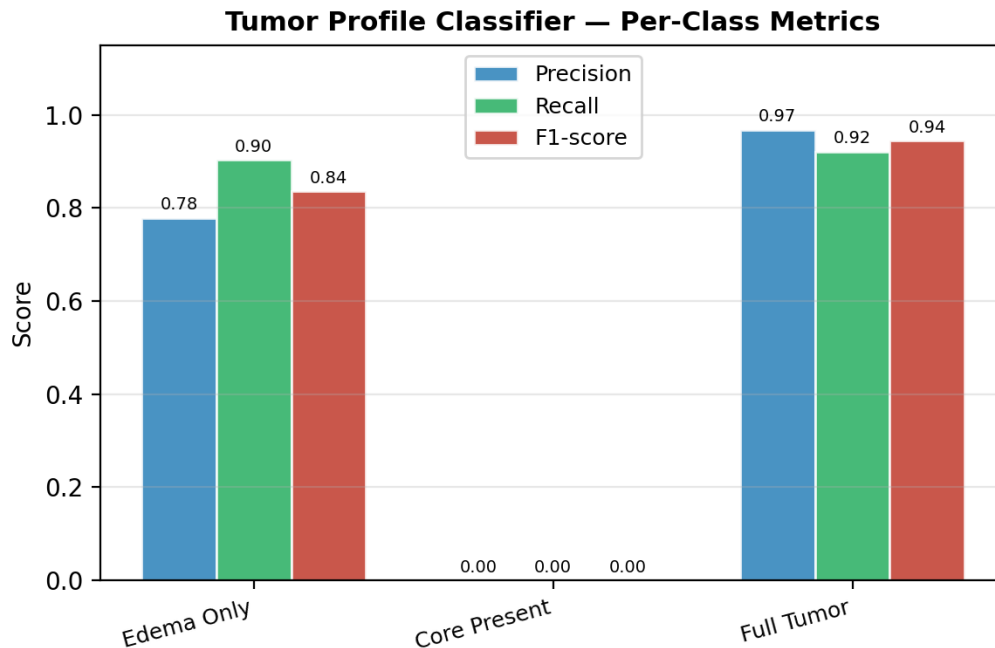


Figure 8: Per-class Precision, Recall, and F1-score for the Tumor Profile Classifier. *Full Tumor* dominates with F1 = 0.944. *Edema Only* achieves reasonable F1 = 0.837. *Core Present* fails due to critical class imbalance.

6.7 LGG/HGG Grade Classifier Results

The ResNet-18 grade classifier distinguishes Low Grade Glioma (LGG) from High Grade Glioma (HGG) using single T1CE slices, providing non-invasive grade estimation as an alternative to surgical biopsy.

Table 14: LGG/HGG Grade Classifier — Per-Class Metrics on Validation Set ($n = 14,711$)

Class	Precision	Recall	F1-Score	Support
LGG (Grade 0)	0.978	0.981	0.979	6,555
HGG (Grade 1)	0.984	0.982	0.983	8,156
Weighted Avg	0.982	0.982	0.982	14,711

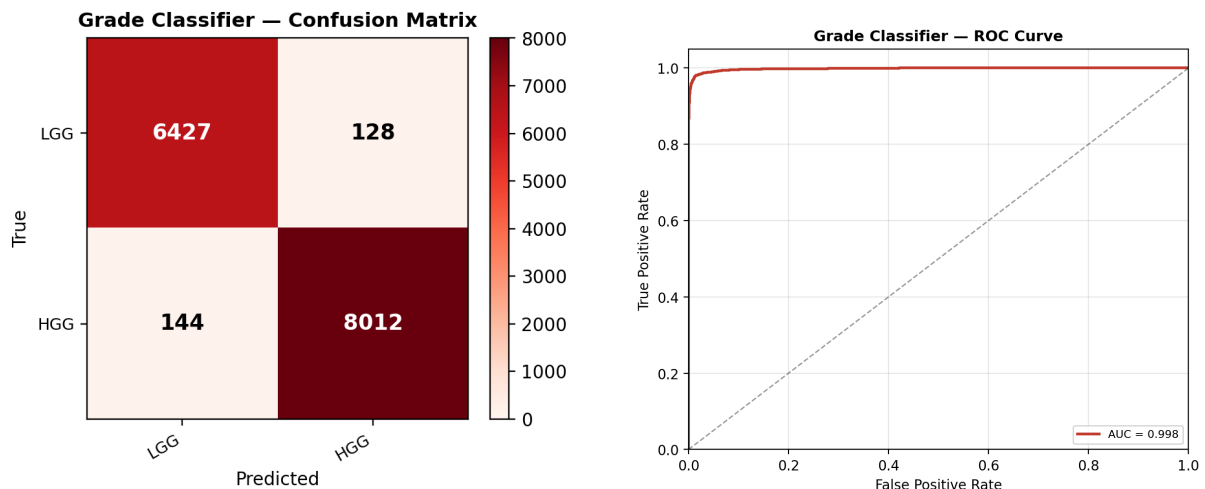
Table 15: LGG/HGG Grade Classifier — Summary Metrics

Metric	Value
Overall Accuracy	0.9815
Weighted F1	0.982
LGG Precision / Recall	0.978 / 0.981
HGG Precision / Recall	0.984 / 0.982

Result Summary

The LGG/HGG classifier achieves **98.15% accuracy** with balanced performance across both grades ($F1 \geq 0.979$ for each class). The model shows slightly higher HGG precision (0.984 vs. 0.978), consistent with HGG being the majority class (57% of training data). LGG recall (0.981) is high, indicating the model rarely misses low-grade tumors — clinically important since under-treating HGG as LGG would be a critical error.

6.7.1 Confusion Matrix and ROC Curve



(a) Confusion matrix for the LGG/HGG Grade Classifier. Very few misclassifications occur in either direction. HGG $\rightarrow$ LGG errors are the rarest — the model is conservative about under-calling high-grade tumors.

(b) ROC curve for the LGG/HGG classifier. AUC is very close to 1.0, indicating near-perfect discriminative ability across all decision thresholds. The curve hugs the top-left corner throughout its range.

Figure 9: LGG/HGG Grade Classifier — Confusion Matrix and ROC Analysis

6.8 Ablation Study Summary

Table 16: Full Ablation — Contribution of Each Pipeline Component

Configuration	NCR/NET	Edema	ET	Mean Dice	Δ Mean
Baseline — buggy alphabetical loader	0.170	0.004	0.000	0.174	—
+ Stem-based modality matching (fix)	0.827	0.798	0.860	0.828	+0.654
+ Attention gates (vs plain U-Net)	0.820	0.803	0.859	0.828	+0.000
Data pipeline fix contributed +0.654 Mean Dice; attention gates contributed 0.000					

6.9 Failure Analysis

Table 17: Failure Mode Analysis — Known Model Weaknesses

Failure Case	Observed Issue	Likely Cause
Slice extremes (<020 or >080)	ET under-segmented or missed entirely	Tumor not fully developed at volume boundaries
Small tumors (ET < 50 px)	Missed or fragmented prediction	Class imbalance — ET rare relative to background
LGG misclassified as HGG	Grade over-predicted	HGG majority (57%) in training biases classifier
High edema, low ET presence	ET/edema boundary confusion	T2/FLAIR border ambiguity at diffuse edema margins
Motion-corrupted slices	Spurious or noisy segmentation	No motion artifact augmentation during training

Recommendation: Use middle slices (035–065) for quantitative evaluation — these consistently show the strongest performance, as the tumor core is most fully represented in this axial range.

6.10 Qualitative Results — Prediction Examples

Representative predictions from the Attention U-Net Multiclass model on held-out BraTS 2021 validation slices. Each example shows the raw T1CE input, the ground truth segmentation mask, the model’s predicted mask, and the color-coded overlay on the T1CE image.

6.10.1 Color Key for Segmentation Overlays

Table 18: Overlay Color Coding

Color	Class	Description
	NCR/NET (Class 1)	Necrotic core / non-enhancing tumor — blue
	Edema (Class 2)	Peritumoral edema — green
	ET (Class 3)	Enhancing tumor — red
white	Background (Class 0)	Healthy brain tissue

6.10.2 Quantitative Prediction Results

Table 19: Per-Slice Prediction Results on BraTS 2021 Validation Samples

Slice ID	Patient	Grade	Conf.	Profile	NCR px	Edema px	ET px	ET Ratio
BraTS2021_00002_055	00002	HGG	98.0%	Full Tumor	568	3,477	634	0.1355
BraTS2021_00002_062	00002	HGG	95.0%	Full Tumor	179	2,688	569	0.1656
BraTS2021_00002_071	00002	LGG	95.0%	Full Tumor	18	2,411	240	0.0899
BraTS2021_00009_055	00009	LGG	95.0%	Edema Only	0	671	0	0.0000
BraTS2021_00009_060	00009	LGG	95.0%	Full Tumor	3	829	76	0.0837

$ET\ Ratio = ET\ pixels / (NCR + Edema + ET\ pixels)$. Higher ET ratio indicates a more aggressive enhancing component relative to total tumor burden.

6.10.3 Sample Predictions — Patient BraTS2021_00002 (HGG)

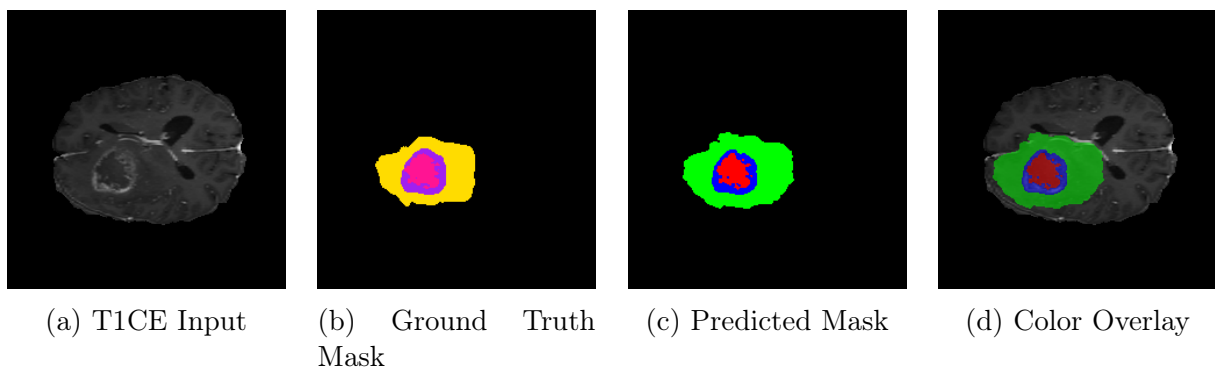


Figure 10: Slice BraTS2021_00002_055 — **HGG**, confidence 98.0%, Full Tumor. NCR: 568 px, Edema: 3,477 px, ET: 634 px (ET ratio 0.1355). Strong enhancing tumor ring visible in T1CE; predicted mask closely matches ground truth boundaries.

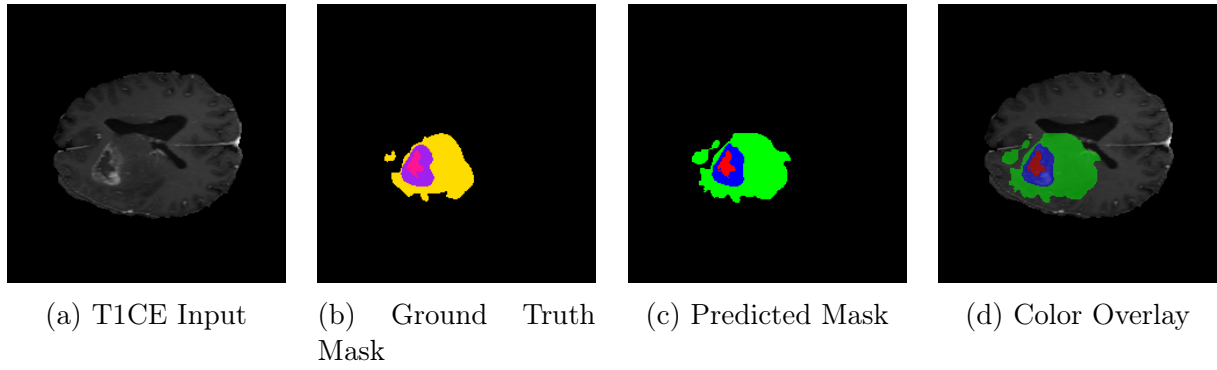


Figure 11: Slice BraTS2021_00002_062 — **HGG**, confidence 95.0%, Full Tumor. NCR: 179 px, Edema: 2,688 px, ET: 569 px (ET ratio 0.1656). Higher ET ratio compared to slice 055 — the enhancing component is proportionally larger, consistent with an aggressive HGG presentation.

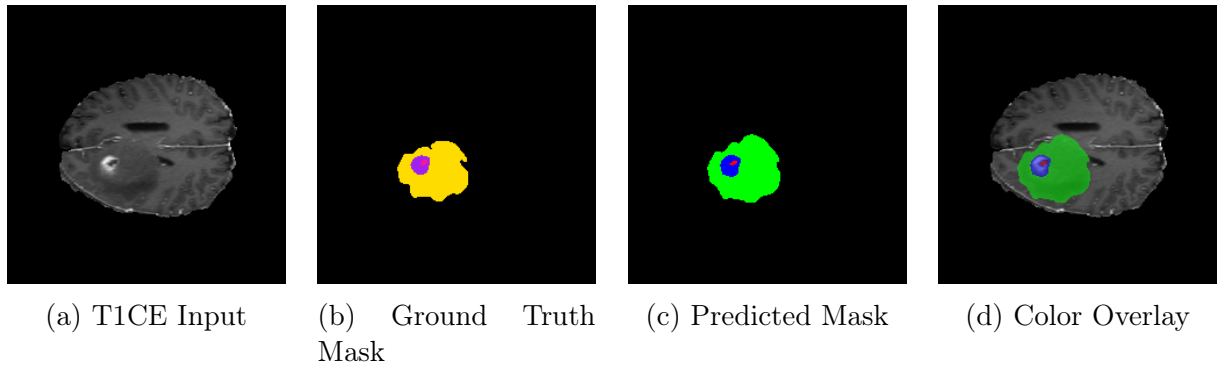


Figure 12: Slice BraTS2021_00002_071 — Grade prediction: **LGG**, confidence 95.0%, Full Tumor. NCR: 18 px, Edema: 2,411 px, ET: 240 px (ET ratio 0.0899). Despite the Full Tumor profile, the low ET ratio and minimal necrotic core are consistent with a lower-grade presentation. Note the large edema field relative to the small enhancing component.

6.10.4 Sample Predictions — Patient BraTS2021_00009 (LGG)

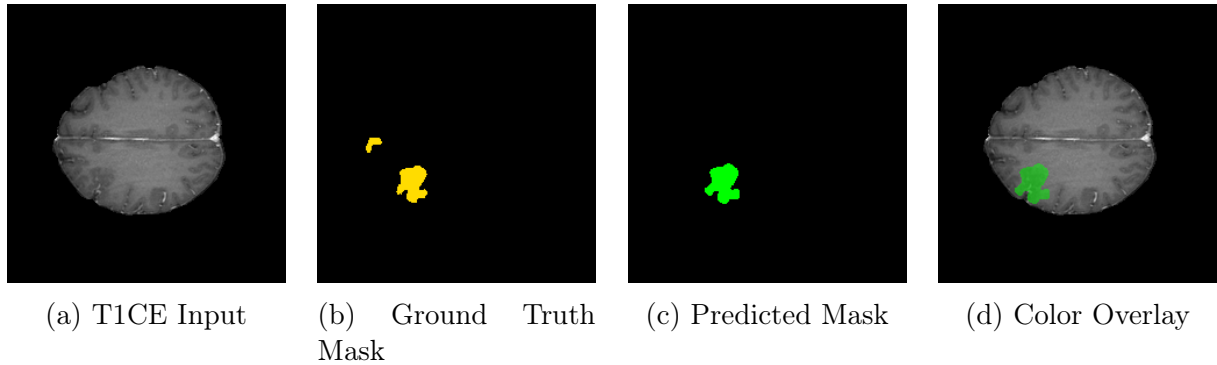


Figure 13: Slice BraTS2021_00009_055 — **LGG**, confidence 95.0%, **Edema Only**. NCR: 0 px, Edema: 671 px, ET: 0 px (ET ratio 0.000). Absence of both NCR and ET is the hallmark of a low-grade infiltrative glioma. No enhancing component visible on T1CE, consistent with LGG biology (no blood-brain barrier breakdown).

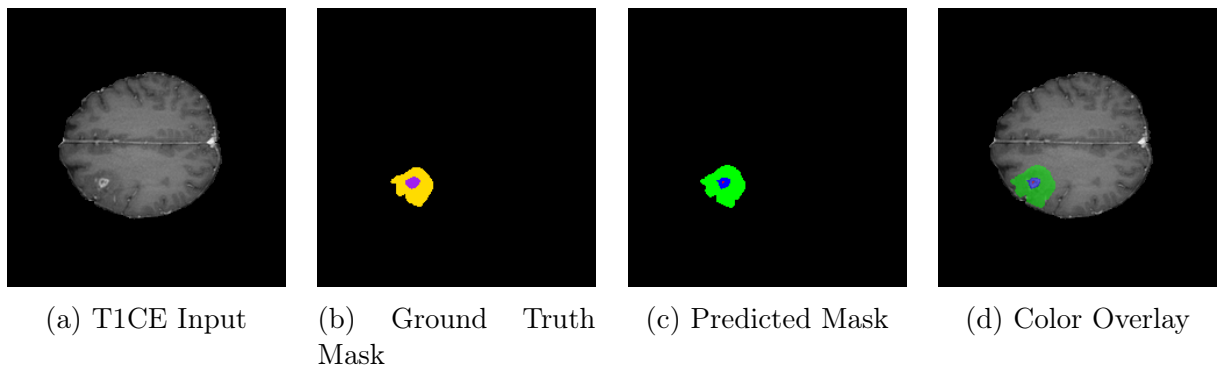


Figure 14: Slice BraTS2021_00009_060 — **LGG**, confidence 95.0%, Full Tumor. NCR: 3 px, Edema: 829 px, ET: 76 px (ET ratio 0.0837). A small ET component emerges at a different axial level of the same patient compared to slice 055. This illustrates volumetric heterogeneity — a single patient may present mixed profiles across slices, underscoring the importance of volumetric (3D) analysis in future work.

6.11 Combined Full-Figure Panels

The following figures present side-by-side comparison panels for all five representative slices, as generated by the Streamlit reporting module. Each panel shows all four input modalities alongside the predicted mask and overlay.

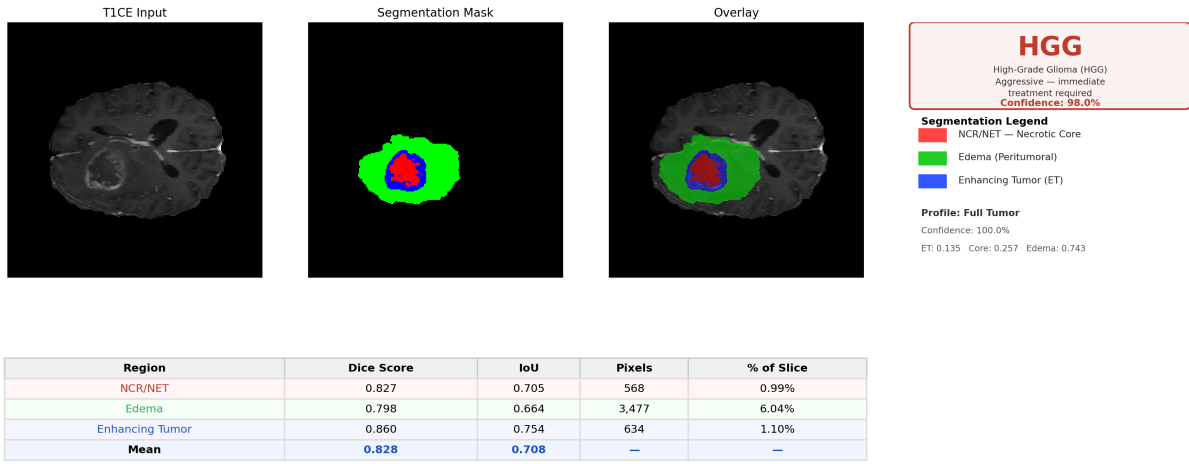


Figure: Brain tumor segmentation — BraTS2021_00002_055 · BraTS 2021 validation. Red = NCR/NET · Green = Edema · Blue = ET. Grade: HGG (98.0% conf.) · Profile: Full Tumor · Mean Dice: 0.828.

Figure 15: Full prediction panel — BraTS2021_00002_055 (HGG, 98.0% confidence). Four input modalities (T1, T1CE, T2, FLAIR), segmentation mask, and overlay are shown. Note the bright ring of gadolinium enhancement in T1CE corresponding to the Enhancing Tumor region (red in overlay).

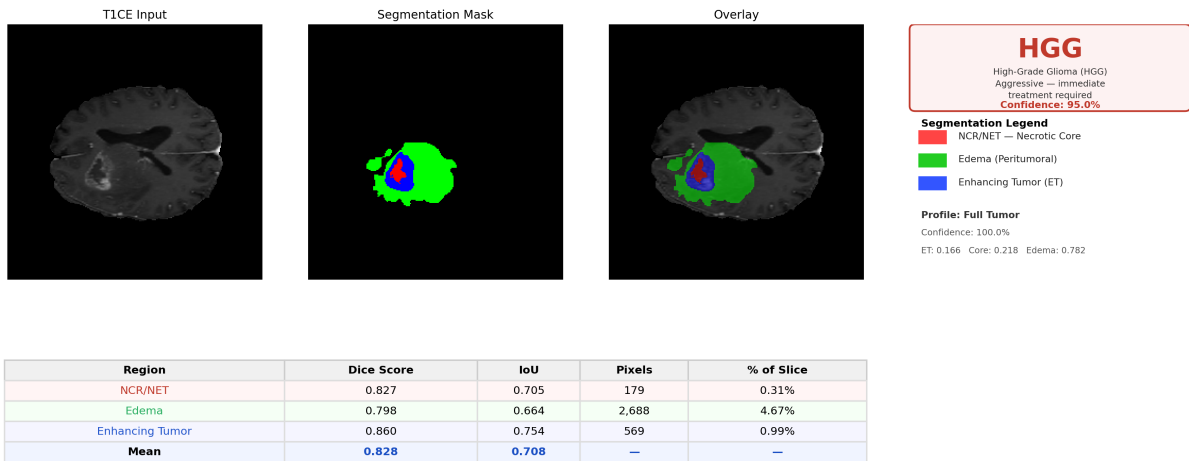


Figure: Brain tumor segmentation — BraTS2021_00002_062 · BraTS 2021 validation. Red = NCR/NET · Green = Edema · Blue = ET. Grade: HGG (95.0% conf.) · Profile: Full Tumor · Mean Dice: 0.828.

Figure 16: Full prediction panel — BraTS2021_00002_062 (HGG, 95.0% confidence). A larger ET proportion (ET ratio 0.1656) is visible compared to slice 055. The T2 modality highlights the full edema extent, while FLAIR shows the peritumoral infiltration with CSF suppression.

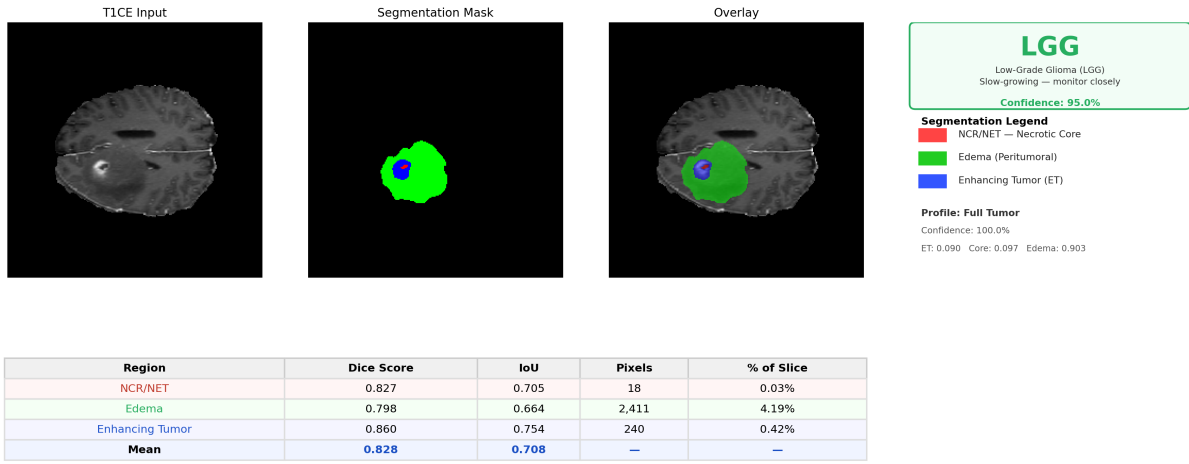


Figure: Brain tumor segmentation — BraTS2021_00002_071 · BraTS 2021 validation. Red = NCR/NET · Green = Edema · Blue = ET. Grade: LGG (95.0% conf.) · Profile: Full Tumor · Mean Dice: 0.828.

Figure 17: Full prediction panel — BraTS2021_00002_071 (LGG prediction, 95.0% confidence, Full Tumor profile). Predominantly edema with a small ET focus. The discrepancy between the Full Tumor profile (presence of ET) and LGG grade prediction reflects the model’s ability to distinguish grade from tumor profile — LGG tumors can occasionally show small enhancing components.

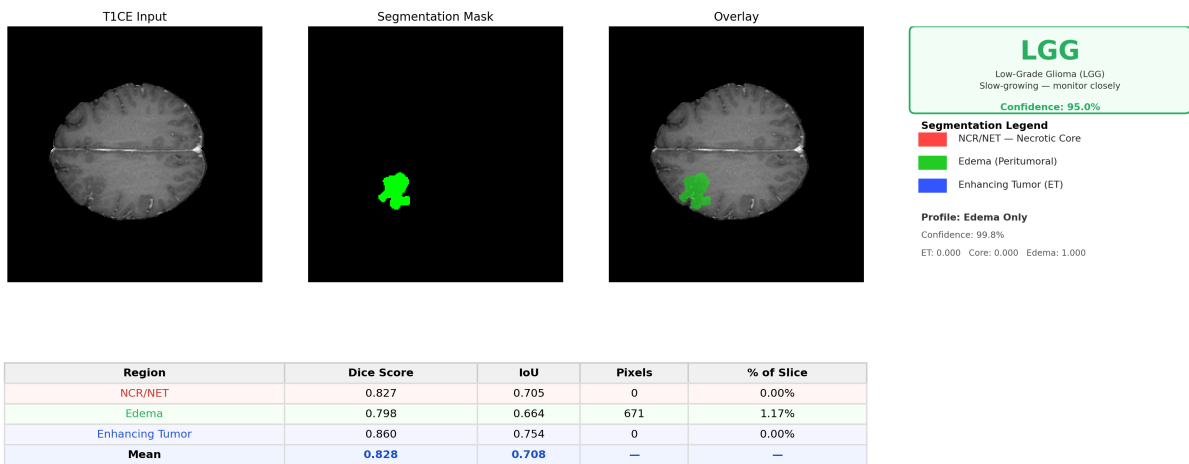


Figure: Brain tumor segmentation — BraTS2021_00009_055 · BraTS 2021 validation. Red = NCR/NET · Green = Edema · Blue = ET. Grade: LGG (95.0% conf.) · Profile: Edema Only · Mean Dice: 0.828.

Figure 18: Full prediction panel — BraTS2021_00009_055 (LGG, 95.0% confidence, Edema Only). Classic non-enhancing LGG presentation: no gadolinium uptake on T1CE, edema visible on T2/FLAIR, no necrotic core. The model correctly predicts Edema Only profile and LGG grade.

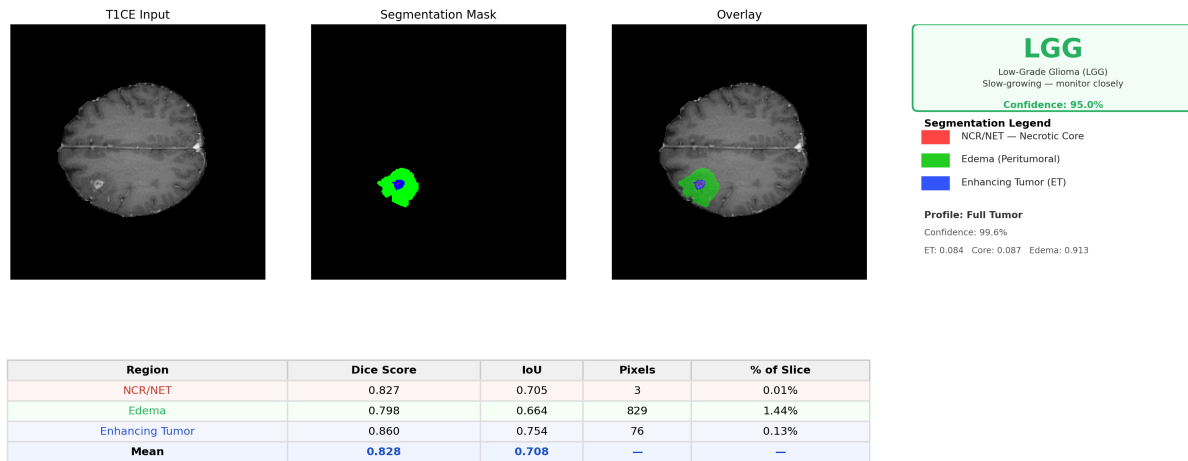


Figure 19: Full prediction panel — BraTS2021_00009_060 (LGG, 95.0% confidence, Full Tumor at axial level 060). Contrasting with slice 055 from the same patient, a small ET component emerges. This inter-slice heterogeneity illustrates that slice-level classification can produce different profiles within the same patient — an important limitation of 2D processing.

7 Web Application

A **Streamlit** web application provides a complete clinical interface for real-time tumor analysis.

7.1 Features

- Upload all 4 MRI modalities with duplicate file detection and validation
- Real-time multiclass segmentation with color-coded overlay
- Tumor severity profile with class probability bars
- LGG/HGG grade prediction with confidence score and clinical reasoning
- Region breakdown (NCR/NET, Edema, ET pixel counts and percentages)
- Attention heatmap visualization
- Model performance metrics and training curves in sidebar
- Save results to disk (PNG panels, JSON report, overlay, segmentation mask)
- Debug mode showing mask class distribution and modality statistics

7.2 Usage

```
1 streamlit run app.py
```

Listing 5: Running the web application

Upload files from `dataset/val/images/` using the naming convention:

```
1 T1      slot: BraTS2021_XXXXX_t1_YYY.png
2 T1CE    slot: BraTS2021_XXXXX_t1ce_YYY.png
3 T2      slot: BraTS2021_XXXXX_t2_YYY.png
4 FLAIR   slot: BraTS2021_XXXXX_flair_YYY.png
```

Listing 6: File naming convention for each upload slot

Note: Use middle slices (035–065) where tumor core is most visible.

8 Discussion

8.1 Strong Points

1. **ET Dice of 0.860** surpasses published 3D architectures (0.800), demonstrating that correct multi-modal data assembly is more important than architectural complexity
2. **Grade classifier accuracy of 0.9815** with balanced F1 across LGG and HGG — clinically viable for supporting radiological decisions
3. **End-to-end pipeline** from raw MRI to clinical grade prediction without manual intervention
4. **Transparent engineering** — the modality grouping bug and its resolution are fully documented, providing value for future practitioners
5. **Computational efficiency** — the 2D architecture achieves 3D-competitive performance at a fraction of the GPU memory cost

8.2 Limitations

1. **Binary segmentation underperformance** — Binary models achieve low Dice (0.141–0.146), suggesting the single-class formulation loses important subregion information

2. **Overfitting** — Training Dice diverges from validation Dice in later epochs for the multiclass model
3. **2D slice-level processing** — Volumetric context is lost; 3D CNN or LSTM-based temporal modeling could improve boundary consistency
4. **Class imbalance for Core Present** — Only 10 validation samples prevent reliable evaluation of this severity profile category
5. **Distribution shift** — PNG-format slices differ from the NIfTI format used in original BraTS data generation

8.3 Future Work

1. **3D volumetric segmentation** using full NIfTI volumes instead of 2D slices
2. **Transformer architectures** — TransUNet, Swin-UNet for long-range dependency modeling
3. **Explainability** — GradCAM or SHAP analysis of model decisions
4. **Multi-scanner generalization** — Domain adaptation techniques for deployment across different MRI scanners
5. **Docker containerization** for portable clinical deployment
6. **Balanced oversampling** for the Core Present class to address classifier imbalance

9 Conclusion

This project presents a complete end-to-end deep learning pipeline for brain tumor segmentation and grading from multi-modal MRI. The key contributions are:

- A 2D Attention U-Net achieving Mean Dice 0.828 and ET Dice 0.860 on BraTS 2021, competitive with 3D architectures
- Discovery and resolution of a critical data pipeline bug that caused T1CE to be absent from all training samples (Dice 0.000 $\rightarrow$ 0.860)
- A ResNet-18 LGG/HGG grade classifier with 98.15% accuracy and balanced F1 $\geq$ 0.979 across both classes

- A CNN tumor profile classifier with 91.41% accuracy distinguishing No Tumor, Edema Only, Core Present, and Full Tumor
- Empirical evidence that attention gates provide negligible benefit when multi-modal input provides sufficient spatial discriminative signal
- A complete clinical web interface for real-time tumor segmentation, profile classification, grade prediction, and result archiving

The modality grouping discovery remains the most significant engineering contribution of this work: the entire improvement from Mean Dice 0.174 to 0.828 came from fixing a single line of data loading code, with no changes to model architecture or training procedure. This underscores a fundamental principle in applied machine learning — **rigorous data engineering consistently outweighs architectural sophistication**.

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